ME-446 · AI & INTERNET OF THINGS · COMPLEX ENGINEERING PROBLEM

Detecting Dangerous Milling Conditions

from vibration and spindle/drive-current statistics

Random forest · 96.2 % recall · MCC 0.762 · grouped validation

Group _____ · NED University of Engineering & Technology

Data: Piecuch & Żabiński, Scientific Data 12, 650 (2025) · CC BY 4.0

A safety monitor for an aggressive cut

The machine does not know which radial depth of cut it is running. The sensor signature does.

The condition

RDOC 8 mm is 80 % of the cutter diameter. Chip load, cutting force and spindle torque rise; chatter and overload become likely.

The risk

A missed dangerous cut means faster wear, scrap or a broken tool. A false alarm costs an inspection, not a spindle.

The task

Binary classification: optimal (4.5 mm) vs non-optimal (8 mm). 120 vibration/current statistics per cycle; no process settings as inputs.

Why it is a genuine CEP problem: RDOC is constant within every tool and experiment. A random row split lets the model memorise tool identity, so the validation design — not just the algorithm — decides whether the result is believable.

968 cycles from a public milling campaign

Piecuch & Żabiński (2025), Scientific Data — processed 120-statistic feature matrix, CC BY 4.0.

968
milling cycles

24
experiments

14
cutting tools

120
sensor statistics

212
dangerous cycles · 21.9 %

2
RDOC levels: 4.5 / 8 mm

- **Two independent sources** feature CSV and metadata workbook cross-checked row for row.
- **Zero missing values** no duplicate rows, no duplicate sensor vectors, no zero-variance features.
- **Difficult records kept** 14 censored cycles and one single-cycle failure retained, not deleted.

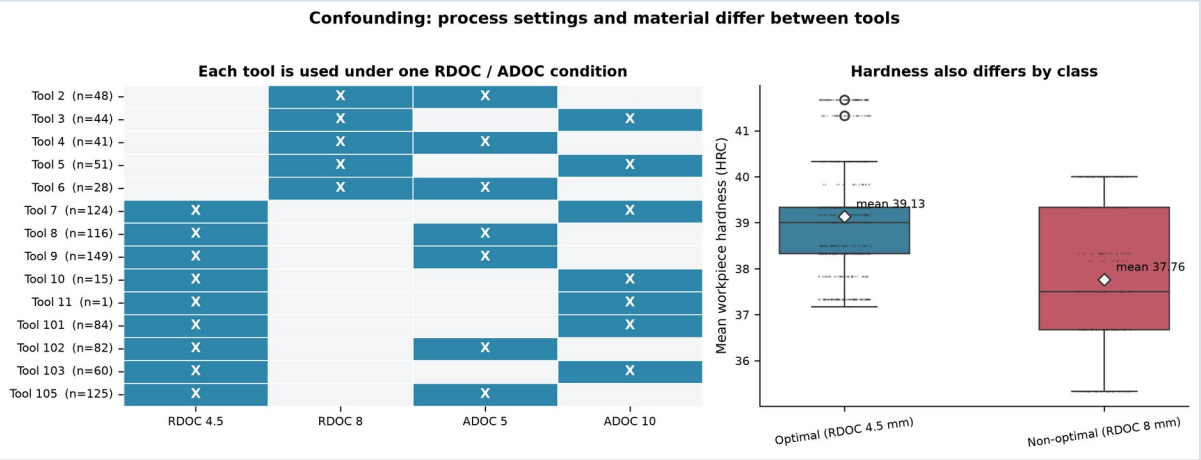
Features the model sees

8 accelerometer channels + 12 current channels, each with min, max, mean, std, skew and kurtosis. Metadata, hardness, wear counters and IDs are excluded.

RDOC is perfectly confounded with tool identity

0 experiments and 0 tools contain both classes — so the split strategy is the result.

- **One condition per tool** Every tool runs at exactly one RDOC/ADOC combination.
- **One condition per experiment** So do all 24 experiments; the class is a property of the setup.
- **Tool type 2 never appears at 8 mm** Material and tool type are entangled with the target.
- **Random split leaks** The same tool appears in train and test; scores become memorisation.



Tool 11 has one cycle; tool type 2 is only ever optimal.

Cleaning → grouped validation → four models

Every model sees the same 120 features and the same folds. Nothing is tuned on the held-out tool.

1

Audit & clean

Parse export quirks; cross-check sources; keep all 968 cycles.

2

Features

signed-log1p on 120 sensor statistics; balanced class weights.

3

Grouped CV

5 tool folds; tuning nested inside each outer fold.

4

Compare

Dummy · logistic · random forest · histogram gradient boosting.

5

Select

Tune threshold on out-of-fold probabilities; stress-test by tool.

Primary estimate

5-fold tool-group CV, pooled out-of-fold predictions. Leave-one-tool-out and leave-one-experiment-out are sensitivity tests.

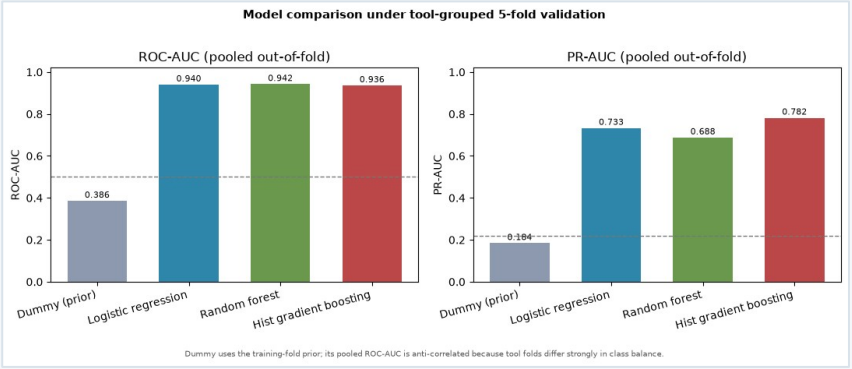
Leakage control: never the headline

Random 5-fold: RF ROC-AUC 0.998. Tool-grouped: 0.942. The gap is the leakage. All reported numbers are grouped.

Four models, one honest comparison

5-fold tool-group CV, threshold 0.5. ROC-AUC and PR-AUC are threshold-independent; the rest are operating-point metrics.

Model	ROC-AUC	PR-AUC	Recall	Precision	F1	MCC	Brier
Dummy (prior)	0.386	0.184	0.000	0.000	0.000	0.000	0.173
Logistic regression	0.940	0.733	0.948	0.663	0.781	0.725	0.093
Random forest	**0.942**	**0.688**	**0.892**	**0.724**	**0.799**	**0.742**	**0.082**
Hist gradient boosting	0.936	0.782	0.877	0.620	0.727	0.650	0.107



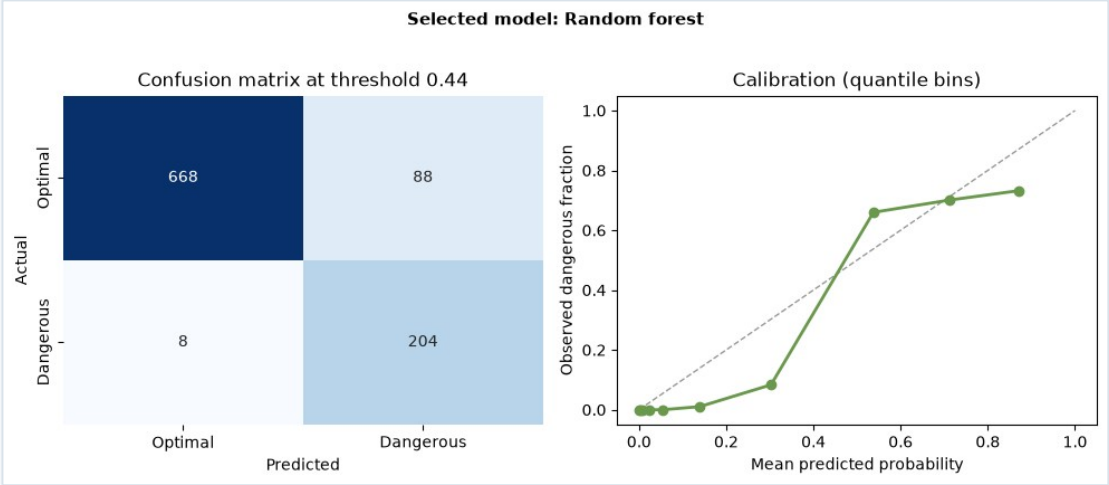
Random forest wins the operating point (F1 0.799, MCC 0.742, Brier 0.082). Gradient boosting wins threshold-independent PR-AUC (0.782) but is weaker at 0.5 — reported honestly, not hidden.

At the tuned operating point, missing a cut is rare

Threshold 0.444 maximises F1 on pooled out-of-fold probabilities; it sits below 0.5 on purpose.

96.2 % recall (dangerous)	**69.9 %** precision	**0.762** MCC
0.923 balanced accuracy	**0.810** F1	**0.082** Brier score

8 missed dangerous cycles out of 212 (3.8 %), 88 false alarms out of 756 (11.6 %). Leave-one-tool-out agrees: recall 0.962, MCC 0.778. The model is not relying on one grouping of tools.



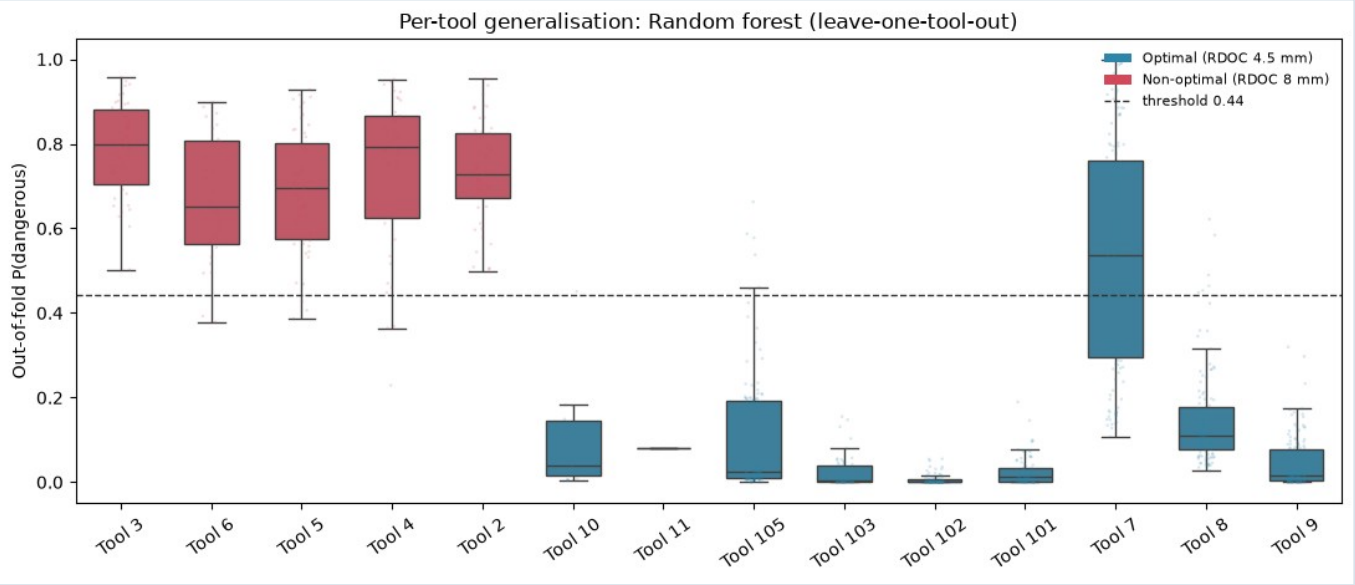
Threshold is a policy: 0.444 favours recall; the dashboard exposes the trade-off.

One optimal tool drives most false alarms

Tool 7 accounts for 75 of the 88 false alarms; the 8 misses are spread across tools 4–6.

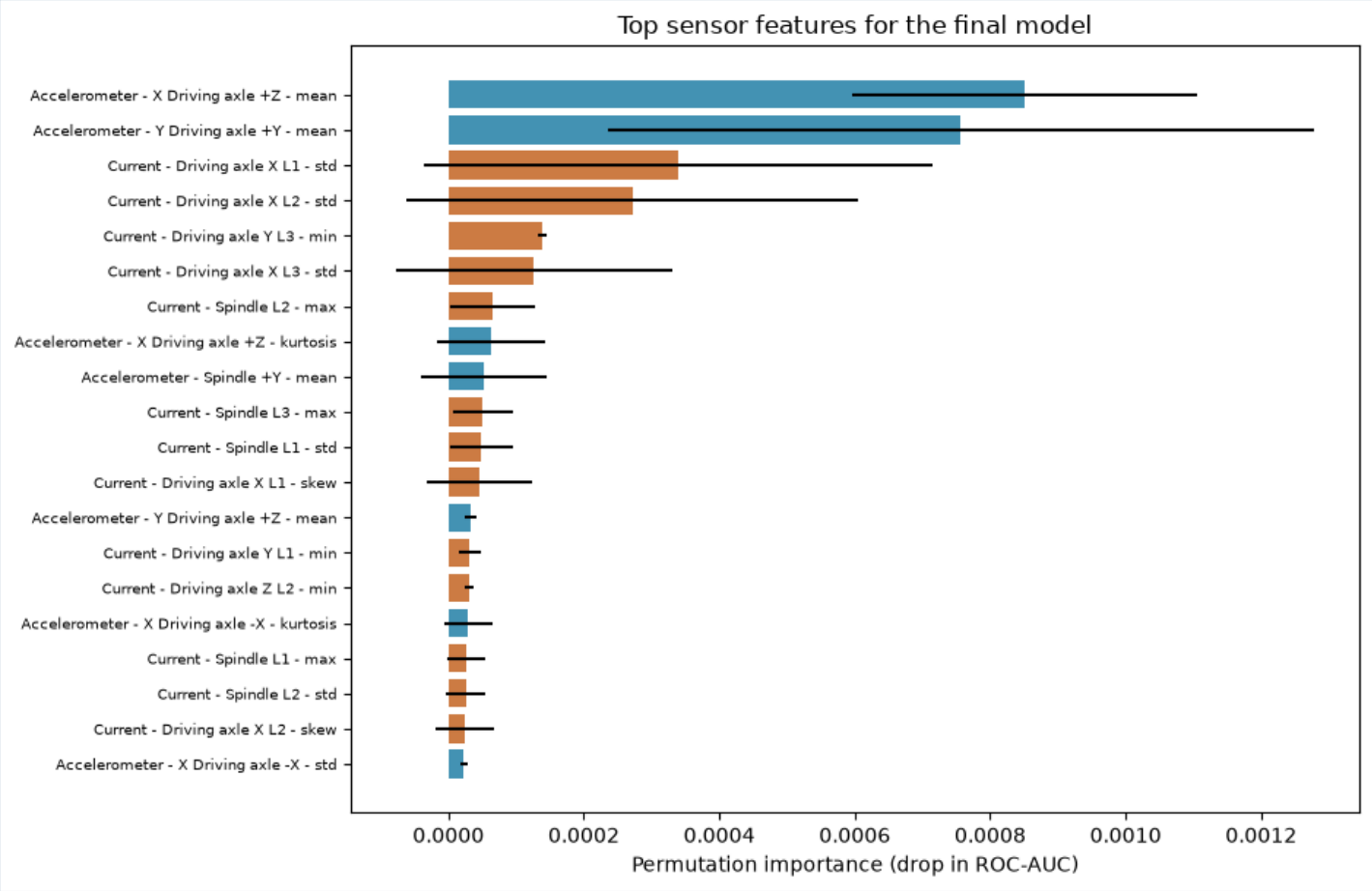
Tool	Actual	Cycles	Mean P	Flagged
4	Dangerous	41	0.712	92.7 %
5	Dangerous	51	0.623	94.1 %
6	Dangerous	28	0.676	92.9 %
7	**Optimal**	**124**	**0.563**	**60.5 %**
8	Optimal	116	0.149	5.2 %
105	Optimal	125	0.104	5.6 %

Tool 7 false alarms cluster late in tool life (mean normalised life 0.33). Wear pushes an optimal tool into a dangerous-looking signature — the central confound.



It reads force, not the answer key

Permutation importance on the fitted forest. Correlated channels share importance; the ranking is the message.



- **Drive-axis mean acceleration** the top signal — a static force offset grows with radial engagement.
- **Current variability** drive and spindle current std follow the heavier, more interrupted cut.
- **No wear counter or ID** metadata is excluded, so the model cannot read the target.
- **Physically plausible** the ranking matches cutting mechanics, not a dataset artefact.

Held-out experiments behave as they should

A fresh forest is trained without P002 and P108, then applied to all 127 of their cycles.

Experiment	Actual	Cycles	Mean P	Flagged
P002	**Dangerous**	**46**	**0.746**	**97.8%**
P108	Optimal	81	0.006	0.0%

The one miss is a real engineering finding

P002's first cycle scores 0.439 — just under the 0.444 threshold. A fresh tool has not yet developed the wear-driven rise in vibration and current. Use a rolling window, not a verdict on one cycle.

Single-cycle scoring with the final model

Cycle	Actual	P(dangerous)	Decision
P002_F12_C4	Dangerous	0.967	Flag
P108_F12_C7	**Optimal**	**0.000**	**Clear**

Same 120-feature interface a deployment would use: one row of sensor statistics, no metadata. P108 is cleared on every cycle; P002 is flagged on 45 of 46.

A data-driven cutting-force monitor

The model's signal and its errors both make physical sense.

SIGNAL

Larger RDOC → higher chip load → higher drive-axis mean acceleration and more variable spindle/drive current.

MODEL

The forest combines correlated channels and shifts its decision boundary at 0.444 to favour recall.

ERROR

Wear, hardness and tool geometry move the signature in the same direction as RDOC, so tool 7 overlaps.

CONSEQUENCE OF AN ERROR

A false negative continues a dangerous cut — faster wear, chatter, scrap or a broken tool. A false positive costs an inspection. The threshold and the human-in-the-loop workflow are chosen around that asymmetry.

Strong screening alarm, not a causal instrument

The confound is a property of the experiment, not something a better model can remove.

LIMITATIONS

- **Confounded design** RDOC constant within each tool and experiment.
- **Wear confound** Dangerous tools fail sooner; signal drifts with life.
- **Small tool count** 14 tools, 5 dangerous; tool 7 dominates errors.
- **Pre-computed features** No raw waveforms; no re-optimised windows.
- **Uncalibrated units** ADC counts, single machine, one campaign.

RECOMMENDATIONS

- **Two-cycle confirmation** Do not retract on one marginal cycle.
- **Tool-specific baseline** Compare with the tool's own early-life signature.
- **Balanced experiment** Cut the same tool at both RDOC levels.
- **Calibrate channels** Convert ADC to g and A before transfer.
- **Tune to cost ratio** Inspection cost vs missed-cut risk.

ME-446 · COMPLEX ENGINEERING PROBLEM

Delivered and reproducible

11 figures	EDA, model comparison, calibration, per-tool evidence
Executed notebook	cleaning → models → threshold → unseen cases
Streamlit dashboard	6 tabs, threshold slider, live scoring, CSV upload
Tableau extracts	8 tables + calculated-field build guide
Technical report	17 pages, 5 400 words, 11 figures

Live dashboard: milling-cep-iguwxgwerngzavkpz2aguj.streamlit.app

github.com/Shahbaz2104/milling-cep · Questions